

Customer Churn Prediction

Objective:

Build a machine learning model to predict whether a customer will leave the company.

Project Tasks / Questions:

Part 1: Data Preprocessing

1. Load the dataset in Python/Jupyter Notebook.
2. Check for null values and duplicates.
3. Handle missing values.
4. Encode categorical variables.
5. Normalize/scale numerical features if required.

Part 2: Exploratory Data Analysis

Answer the following:

1. What percentage of customers have churned?
2. Which contract type has the highest churn?
3. Does monthly charge impact churn?
4. Which features correlate most with churn?

Part 3: Model Building

Build at least 2 ML models such as:

- Logistic Regression
- Random Forest
- Decision Tree
- XGBoost

Part 4: Model Evaluation

Evaluate using:

- Accuracy
- Precision
- Recall

- F1 Score
- Confusion Matrix

Part 5: Business Recommendations

Suggest strategies to reduce churn.

Deliverables:

- ✓ Python code / Jupyter Notebook
- ✓ Trained Model
- ✓ Visualizations
- ✓ Project Repo